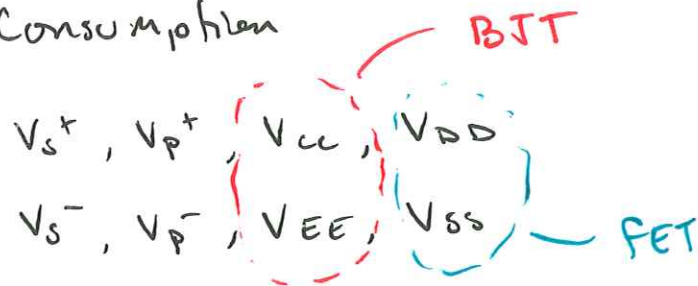
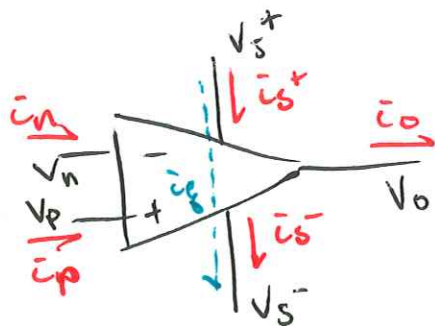


Op Amp Powering & Current Consumption



Baseline Case: $V_n = V_p = 0$ (or some other $\neq 0$) $\rightarrow V_o = 0$
 then consider R_L on V_o to GND $\rightarrow i_o = 0$

and $i_n \approx i_p \approx 0$

$$i_{s+} = i_{s-} \neq 0 \rightarrow i_{s+} = i_{s-} = i_q \quad i_q = \text{quiescent current.}$$

Now Consider $V_o \neq 0 \therefore i_o \neq 0$

i_{s+} & i_{s-} WILL ALWAYS BE POSITIVE

$\rightarrow V_{s+}$ only source current
 V_{s-} only sink current

if $i_o > 0$, then

$$i_{s+} = i_q + i_o$$

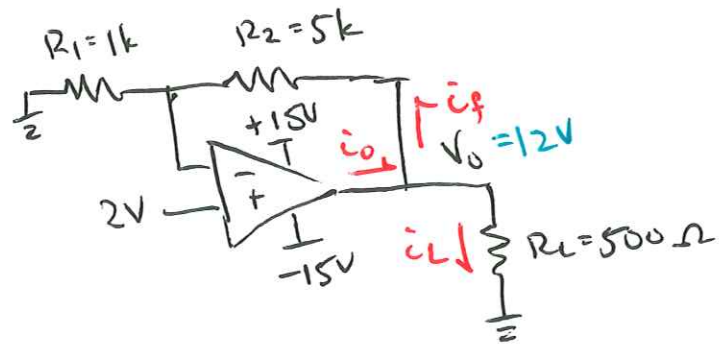
$$i_{s-} = i_q$$

if $i_o < 0$, then

$$i_{s+} = i_q$$

$$i_{s-} = i_q - i_o$$

Powering Example:



$$i_n = i_p = 0$$

ease of analysis: $r_d \rightarrow \infty$, $r_o \rightarrow 0$, $\alpha \rightarrow \infty$

$$A_{ideal} = 1 + \frac{R_2}{R_1} = 6 \text{ V/V}$$

$$V_o = 12 \text{ V}$$

$$i_o = i_f + i_L = \frac{V_o - V_i}{5k} + \frac{12V}{500\Omega} = 2\text{mA} + 24\text{mA}$$

$$i_o = 26\text{mA}$$

What is Power Consumed?

Power via output: $P_o = VI$

$$P_o = (15V - 12V) \cdot 26\text{mA} = 78\text{mW}$$

i_o is current from V_{s+} to V_o

Power via i_g

$$P_g = (15V - (-15V)) i_g$$

$$P_g = 90\text{mW}$$

i_g from Datasheet
ex. OP27: $i_g = 3\text{mA}$

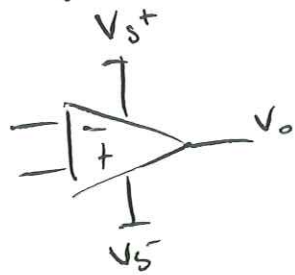
total

$$P_T = P_o + P_g = 168\text{mW}$$

max Power from datasheet: $P = 140\text{mW}$

over power!

Voltage Output Range Limitations



Ideal case, $V_{S-} \leq V_O \leq V_{S+}$

Actual case: $V_{S-} + X^- \leq V_O \leq V_{S+} - X^+$

$\downarrow$ $\downarrow$
 V_{OL} V_{OH}

where X^- & X^+ are some margins to the supplies.

X^- & X^+ usually $\approx$ equal.

V_{OL} & V_{OH} commonly stated as "OVS" or "output voltage swing"

OVS usually = $V_{OH} - V_{OL}$ → assume these are symmetric!
 but not necessarily

ex. OP27 if $V^+/V^- = \pm 15V \rightarrow OVS = \pm 13.5V$
 $\rightarrow V_{OL} = -13.5V$
 $V_{OH} = 13.5V$

ex. ~~AD4807~~ ^{??} if $V^+/V^- = \pm 15V \rightarrow OVS = 27V$
 $\rightarrow V_{OL} = -OVS/2 = -13.5V$
 $V_{OH} = OVS/2 = 13.5V$

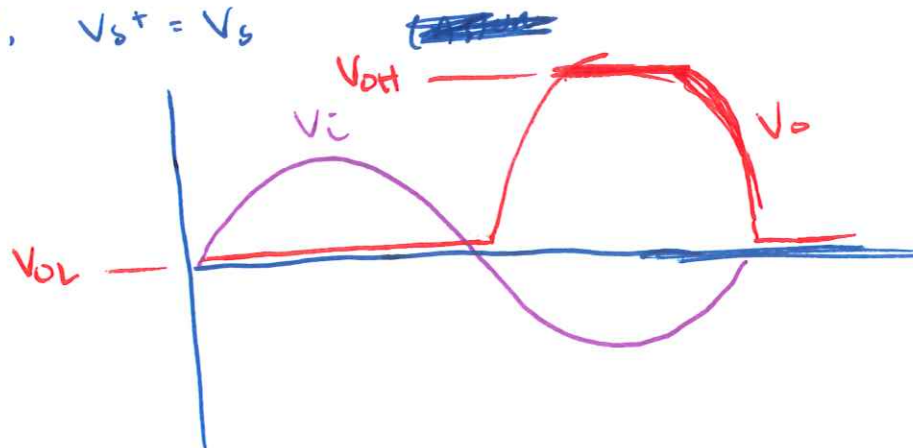
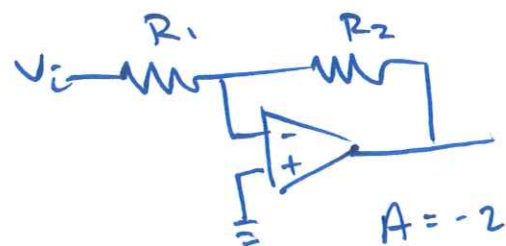
Rail-to-Rail Amplifiers:

Op-Amp has $V_{OL} \approx V_{S-}$, $V_{OH} \approx V_{S+}$

ex AD4807 $\rightarrow V_{OH} = V_{P+} - 0.04V$
 $V_{OL} = V_{P-} + 0.07V$

Dealing with Single-Ended Supplies.

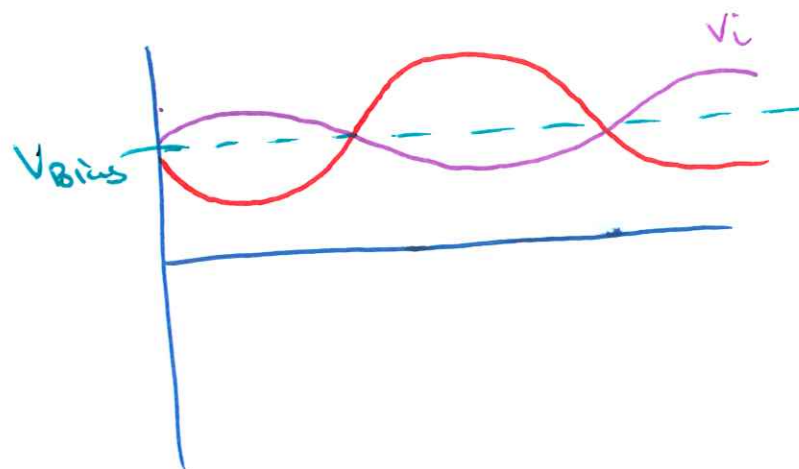
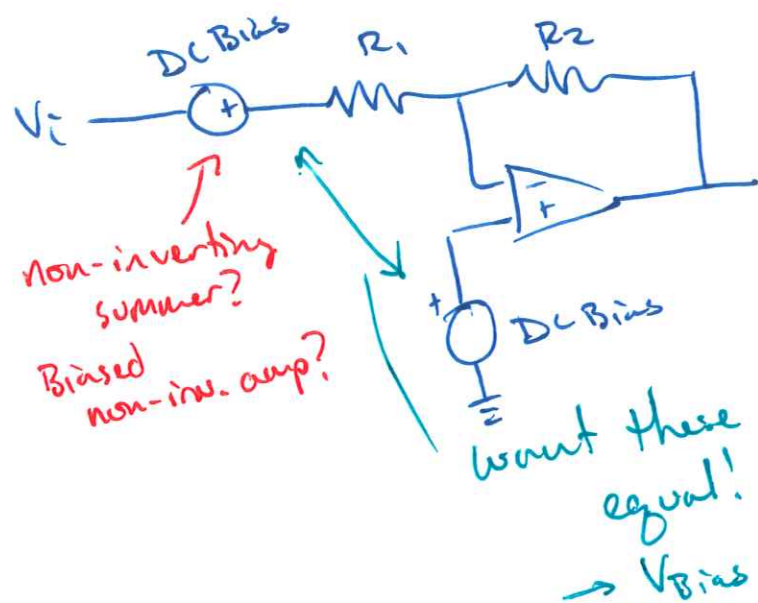
that is $V_S^- = 0$, $V_S^+ = V_S$



fix by DC Biasing

→ Add DC signal to input

→ Cause Amplifier to invert around a DC value $\neq 0$



typically $V_{bias} = V_S^+ / 2$

General types of op-Amp circuits:

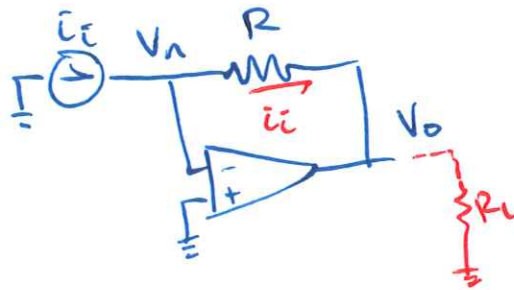
$$V \rightarrow V, \quad V \rightarrow \underline{I}, \quad I \rightarrow V, \quad I \rightarrow \underline{I}$$

$V = \text{voltage}$
 $I = \text{current}$

$$V \rightarrow V: A = \frac{V_o}{V_i} \text{ or } = \frac{V_o \leftarrow V_{\text{bias}}}{V_i \text{ ~~not~~ ~~source~~}} \quad \text{unit: } \frac{V}{V} = 1$$

$$I \rightarrow V: A = \frac{V_o}{\underline{I}_i} \quad \text{unit } \frac{V}{A} = \Omega$$

"trans resistance" Amp



By virtual short, $V_n = 0$

$$V_n - V_o = i_i R \rightarrow \frac{V_o}{i_i} = -R$$

ideal case:

~~if~~ V_n a perfect ground, then $R_i = 0 \Omega$

Topology:

$$\begin{aligned} \text{shunt input} &\rightarrow R_i \approx \frac{R}{1+A} \approx \frac{R}{1+L} \rightarrow 0 \Omega \quad \text{if } L \text{ large} \\ \text{shunt output} &\rightarrow R_o \approx \frac{r_o}{1+L} \rightarrow 0 \Omega \quad \text{if } L \text{ large.} \end{aligned}$$

→
 next page
 example when
 L not large

simple I-V conv. Input R issues.

Assume ~~$r_o \gg R$~~ , $r_o \gg R$, $r_o \approx 75 \Omega$, $a = 10^4 \text{ V/V}$
 $r_o \rightarrow \infty$

$$A_{ideal} = -1000 \text{ V/A} = -1 \text{ V/mA}$$

$$\text{then } R = 1 \text{ k}\Omega$$

$$\text{using } R_i = R/(1+a),$$

$$R_i = 0.1 \Omega$$

$$A_{ideal} = -100000 = -0.1 \text{ V/nA} \rightarrow R = 100 \text{ k}\Omega$$

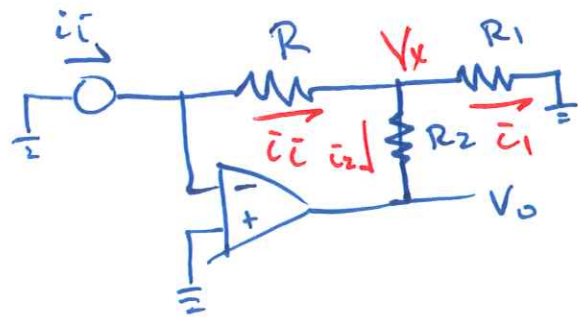
$$R_i = 10 \Omega$$

$$A_{ideal} = -0.1 \text{ V/nA} \rightarrow R = 100 \text{ M}\Omega$$

$$R_i = 10 \text{ k}\Omega \leftarrow \text{Not an ideal current sink!}$$

Also: $100 \text{ M}\Omega$ is crazy large
(bad AC noise!)

Reduce Required R for high gain I-V applications.



Assume $a \rightarrow \infty$

$$\bar{i}_i = \bar{i}_2 + \bar{i}_1 = \frac{0 - V_x}{R} = \frac{V_x - V_o}{R_2} + \frac{V_x}{R_1}$$

$$A = \frac{V_o}{\bar{i}_i} = -R \left(1 + \underbrace{\frac{R_2}{R} + \frac{R_2}{R_1}}_k \right) = -kR$$

method for picking R, R_1, R_2 :

- Pick reasonably valued R

$$R = 1 \text{ M}\Omega$$

- Pick value for R_1 or R_2 and solve for other.

$$R_1 = 1 \text{ k}\Omega$$

$$\text{for } A = -0.1 \text{ V/nA} \rightarrow R_2 = 100 \text{ k}\Omega$$

Still ends up that $R_2 \approx 10 \text{ k}\Omega$

Does not improve current sinking!